

Aligning Metabolomic Networks via Graph Learning for Common Subnetwork Discovery

Edwin A.M.¹, Deybis Paucar Pinto¹, Alfredo J. Ibañez¹, Madina Mansurova¹



June, 2026

1 Introduction

2 Methodology

3 Experiments

4 Results

5 Discussion

6 Conclusion

1 Introduction

2 Methodology

3 Experiments

4 Results

5 Discussion

6 Conclusion

Problem definition

Let two graphs be defined as $G_1 = (V_1, E_1)$, $G_2 = (V_2, E_2)$.

The node alignment problem consists of finding a mapping $\pi : V_1 \rightarrow V_2$, such that each node $v \in V_1$ is assigned to a corresponding node $\pi(v) \in V_2$.

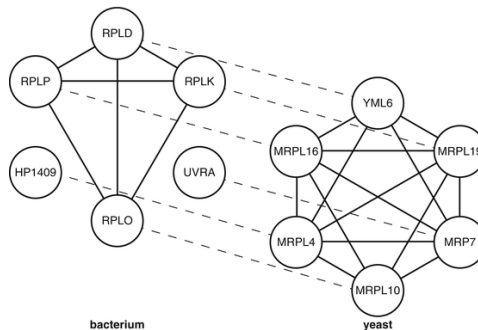


Figure 1: Node alignment.

Source: https://link.springer.com/rwe/10.1007/978-1-4419-9863-7_992

MS dataset format

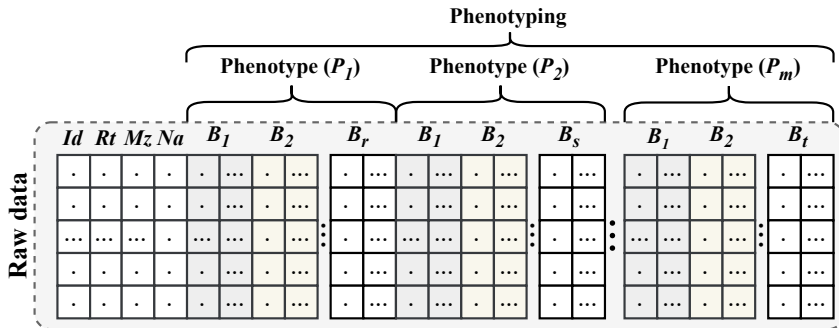


Figure 2: Aligned mass spectrometry data format.

Base GNN model

T-GAE: Transferable Graph Autoencoder for Network Alignment.
(He, J., Kanatsoulis, C., & Ribeiro, A., 2024)



Figure 3: T-GAE encoder.

1 Introduction

2 Methodology

3 Experiments

4 Results

5 Discussion

6 Conclusion

Methodology

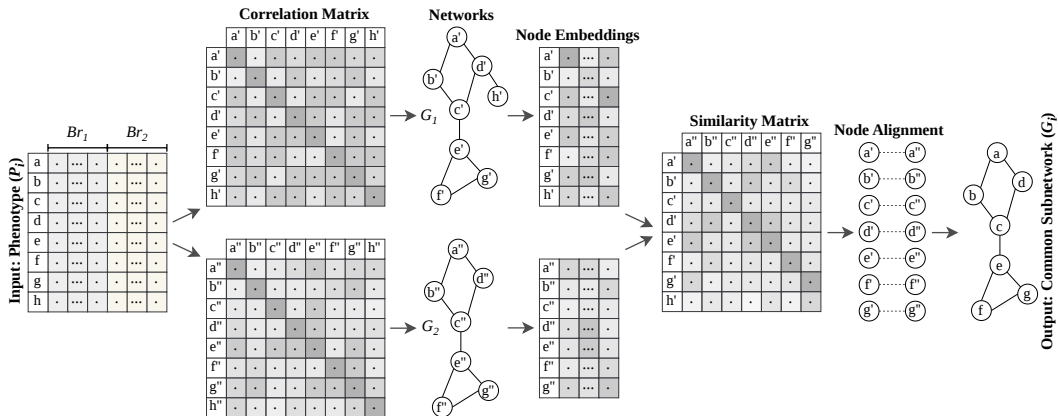


Figure 4: Our workflow.

1 Introduction

2 Methodology

3 Experiments

4 Results

5 Discussion

6 Conclusion

1 Introduction

2 Methodology

3 Experiments

4 Results

5 Discussion

6 Conclusion

Build metabolomic networks

Table 1: Dataset information.

Group	Subgroup	$ V $	$ E $	Density	Connected
<i>secoAmazonas</i>	1	2071	1 113 277	0.52	True
<i>secoAmazonas</i>	2	2071	1 085 089	0.51	True
<i>secoSanMartin</i>	1	2069	1 082 335	0.51	True
<i>secoSanMartin</i>	2	2071	1 103 292	0.51	True
<i>secoCusco</i>	1	2072	1 091 447	0.51	True
<i>secoCusco</i>	2	2072	1 086 182	0.51	True
<i>frescoCusco</i>	1	2071	1 114 859	0.52	True
<i>frescoCusco</i>	2	2072	1 089 225	0.51	True

Node embedding (T-GAE)

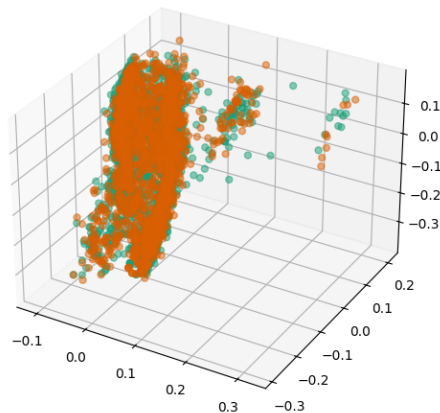


Figure 5: Node embedding (*secoSanMartin*).

Network alignment

Table 2: Node alignment (*secoSanMartin*).

ID 1	ID 2
13	13
19	252
27	378
31	1148
38	38
43	273
49	72
50	50
51	366
53	241
73	270
...	...
1825	1726

Table 3: Node filtered **133/2072** (*secoSanMartin*).

ID	Rt	Mz	Metabolite name
13	6.10	281.05	Unknown
38	6.16	182.10	Unknown
50	6.20	354.10	Unknown
128	6.47	119.20	Unknown
184	6.76	341.14	Unknown
261	7.23	371.06	Unknown
272	7.29	327.01	Unknown
314	7.61	57.02	DECANE; EI-B; MS
336	7.77	53.96	Unknown
339	7.81	328.01	Unknown
400	8.31	109.08	CITRONELLYL ACETATE...
402	8.31	109.13	BETA-HOMOCYCLOCITRAL...
407	8.37	401.11	Unknown

1 Introduction

2 Methodology

3 Experiments

4 Results

5 Discussion

6 Conclusion

1 Introduction

2 Methodology

3 Experiments

4 Results

5 Discussion

6 Conclusion

Thanks!
∞

edwin.alvarez@pucp.edu.pe
<https://github.com/win7>

- Erciyes, K. (2018). *Guide to graph algorithms*
- Yao Ma, Jiliang Tang. (2021). *Deep Learning on Graphs*.
- Stamile C., Marzullo A., Deusebio E. (2021). *Graph Machine Learning*.
- Wu, L., Cui, P., Pei, J., Zhao, L., & Guo, X. (2022). *Graph neural networks: foundation, frontiers and applications*.
- Knickerbocker, D. (2023). *Network Science with Python*.
- Labonne, M. (2023). *Hands-On Graph Neural Networks Using Python*.
- Broadwater, K., & Stillman, N. (2025). *Graph neural networks in action*.